

SUPPLY CHAIN NETWORK DESIGN & IMPLEMENTATION

Multi-Site Network with VLANs, OSPF, NAT/PAT, DHCP and ACL Security

Prepared by: Ziad Waleed Fathy

Project: Supply Chain Network

Tool: Cisco Packet Tracer

Date: September 2026

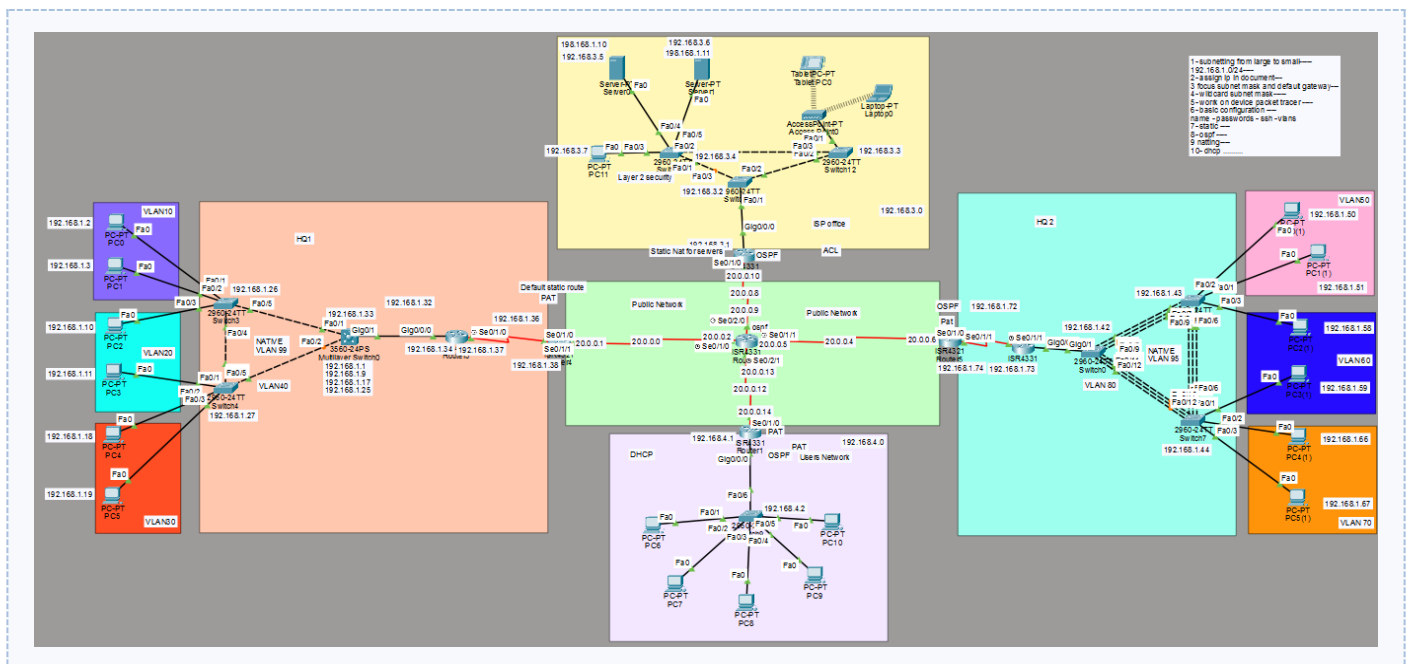


Figure 0 – Complete Network Topology

1. Project Overview

This project presents the design and implementation of a multi-site enterprise network using Cisco Packet Tracer. The network consists of multiple sites connected through a central core router and public WAN links.

The implementation includes VLAN segmentation, subnetting, Inter-VLAN routing, OSPF dynamic routing, static routing, NAT, PAT, DHCP, and ACL-based traffic control.

The main objective is to provide controlled connectivity between all sites and the centralized servers while preventing direct communication between different sites.

2. Project Objectives

Objectives

1. Design a scalable multi-site enterprise network.
2. Perform subnetting using VLSM.
3. Implement VLAN segmentation.
4. Configure Inter-VLAN Routing.
5. Configure OSPF for dynamic routing.
6. Configure Static Routing where required.
7. Implement PAT for private networks.
8. Implement Static NAT for public servers.
9. Configure DHCP for the bottom site.
10. Implement ACLs for traffic control.
11. Allow all sites to access the central servers.
12. Prevent communication between different sites.
13. Verify the entire network using Ping and Cisco verification commands.
14. Remote Access — Telnet & SSH

3. Network Topology

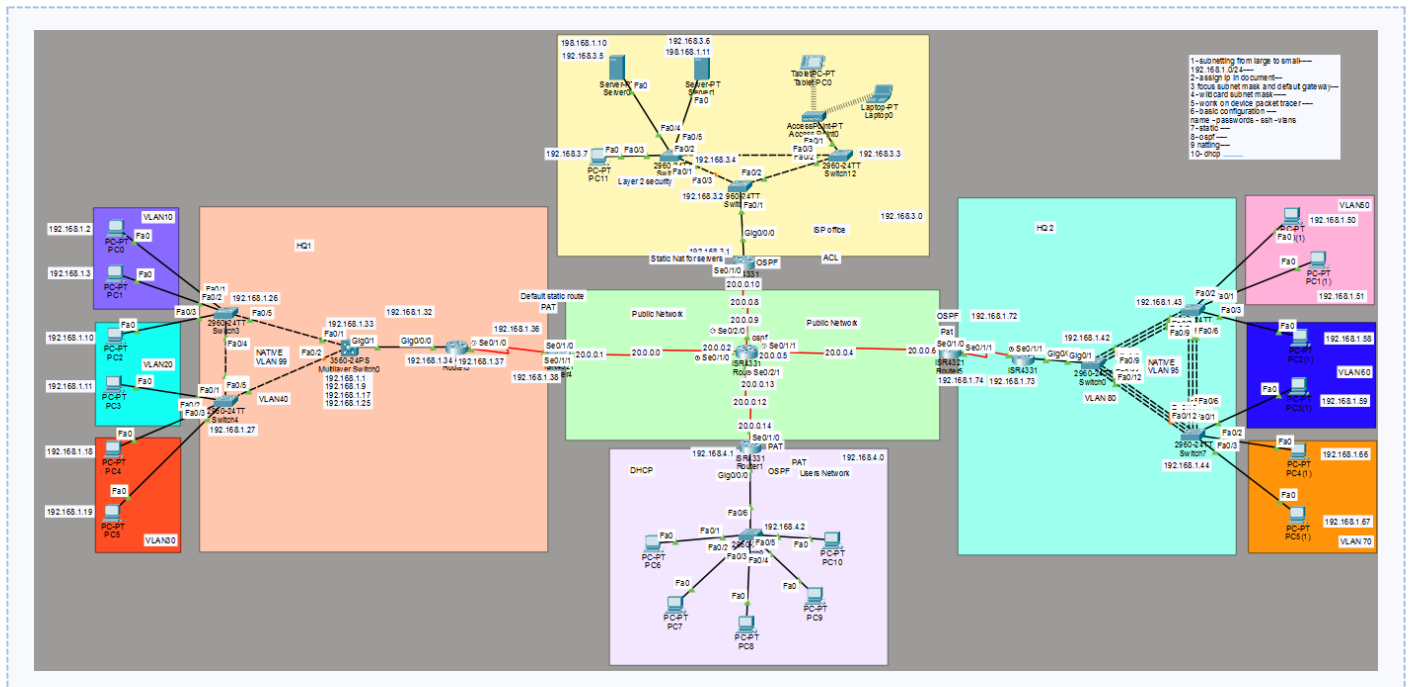
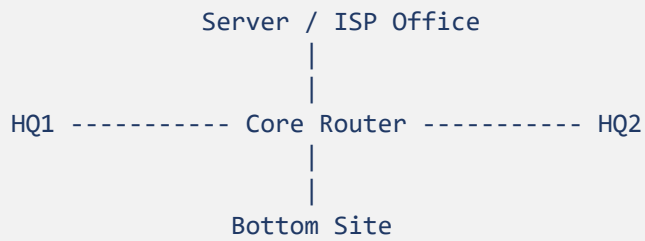


Figure 1 – Complete Enterprise Network Topology

The network consists of:

- HQ1
- HQ2
- Server / ISP Office
- Bottom / User Network
- Central Core Router
- Public WAN Network

4. Network Architecture



Site	Main Technologies
HQ1	VLANs, Inter-VLAN Routing, PAT, Static Default Route
HQ2	VLANs, Inter-VLAN Routing, OSPF, PAT
Server Site	Static NAT, OSPF
Bottom Site	DHCP, PAT, OSPF
Core	OSPF, Public WAN

5. Network Requirements

Functional Requirements

15. Each site must be subnetted independently.
16. Each VLAN must have its own subnet.
17. All sites must reach the central servers.
18. Devices inside the same site must communicate.
19. Different sites must not communicate directly.
20. Server addresses must be reachable using public IP addresses.
21. Private users must use PAT.
22. The bottom site must obtain IP addresses using DHCP.
23. OSPF must be used on the specified routers.
24. HQ1 must use a default static route.

6. IP Addressing & Subnetting

6.1 Private Addressing

Each site is subnetted independently, beginning from the same private base network 192.168.1.0/24. This design choice highlights that each site's addressing scheme is self-contained and does not depend on the addressing of any other site.

Site	Base Network
HQ1	192.168.1.0/29
HQ2	192.168.1.40/29
ISP/Server	192.168.3.0/28
Bottom	192.168.4.0/28
MLS-Router3	192.168.1.32/32
Router3-Router4	192.168.1.36/32
Router5-Router7	192.168.1.72/32
Router7	192.168.1.(41-49-57-65)

7. VLSM Subnetting

HQ1

VLAN	Network	Mask	Gateway
VLAN 10	192.168.1.0/29	255.255.255.248	192.168.1.1
VLAN 20	192.168.1.8/29	255.255.255.248	192.168.1.9
VLAN 30	192.168.1.16/29	255.255.255.248	192.168.1.17
VLAN 40	192.168.1.24/29	255.255.255.248	192.168.1.25

HQ2

VLAN	Network	Mask	Gateway
VLAN 50	192.168.1.48/29	255.255.255.248	192.168.1.49
VLAN 60	192.168.1.56/29	255.255.255.248	192.168.1.57
VLAN 70	192.168.1.64/29	255.255.255.248	192.168.1.65
VLAN 80	192.168.1.40/29	255.255.255.248	192.168.1.41

Bottom

Network: 192.168.4.0/28

Gateway: 192.168.4.1

Server Network

Network: 192.168.3.0/28

Gateway: 192.168.3.1

8. WAN / Public IP Addressing

Connection	Network	Side A	Side B
HQ1 ↔ Core	20.0.0.0/30	.1	.2
Core ↔ HQ2	20.0.0.4/30	.5	.6
Core ↔ Server	20.0.0.8/30	.9	.10
Core ↔ Bottom	20.0.0.12/30	.13	.14

The 20.0.0.0/30 networks are used as public WAN point-to-point links between the core router and the different sites.

9. VLAN Design

Site	VLAN	Purpose
HQ1	10	Users
HQ1	20	Users
HQ1	30	Users
HQ1	40	Users
HQ1	99	Native VLAN
HQ2	50	Users
HQ2	60	Users
HQ2	70	Users
HQ2	80	Users
HQ2	90	Native VLAN

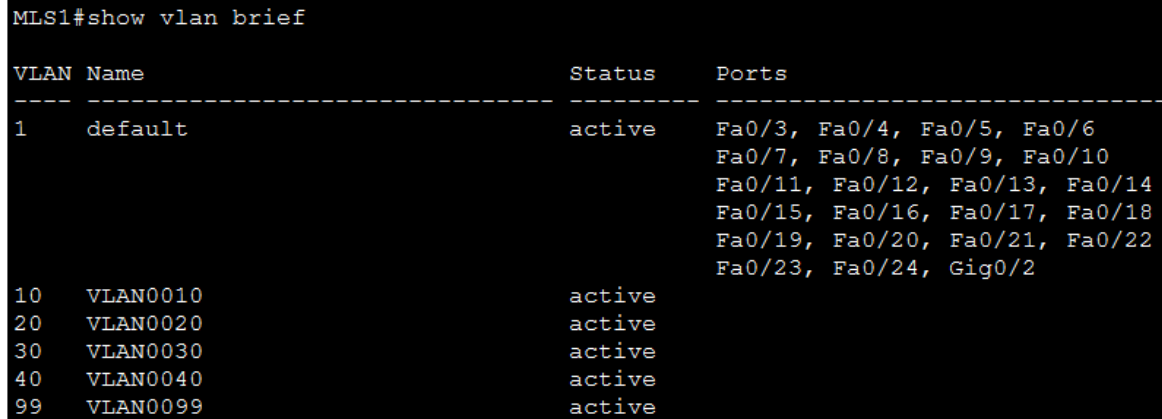
10. VLAN Configuration

10.1 HQ1 VLAN Verification

Objective: Verify that all required VLANs have been created on the HQ1 switch.

Command:

```
show vlan brief
```



```
MLS1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	VLAN0010	active	
20	VLAN0020	active	
30	VLAN0030	active	
40	VLAN0040	active	
99	VLAN0099	active	

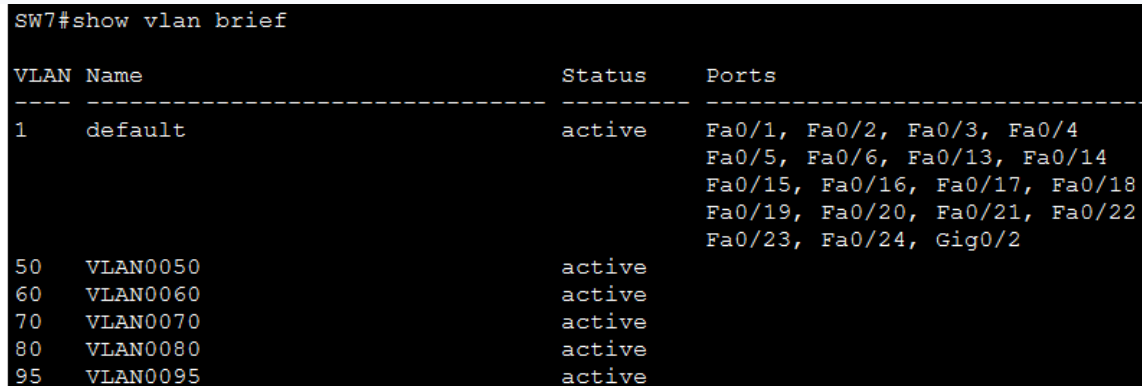
Figure 2 – HQ1 VLAN Configuration

Result: All HQ1 VLANs (10, 20, 30, 40, 99) are present and correctly assigned to their respective ports.

10.2 HQ2 VLAN Verification

Objective: Verify that all required VLANs have been created on the HQ2 switch.

```
show vlan brief
```



```
SW7#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
50	VLAN0050	active	
60	VLAN0060	active	
70	VLAN0070	active	
80	VLAN0080	active	
95	VLAN0095	active	

Figure 3 – HQ2 VLAN Configuration

Result: All HQ2 VLANs (50, 60, 70, 80, 95) are present and correctly assigned to their respective ports.

11. Trunk Configuration

11.1 HQ1 Trunk Verification

Objective: Confirm that trunk links are correctly established and carrying the expected VLANs.

```
show interfaces trunk
```

```
MLS1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     auto      n-802.1q       trunking    99
Fa0/2     auto      n-802.1q       trunking    99

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,99
Fa0/2     none
```

Figure 4 – HQ1 Trunk Configuration

11.2 HQ2 Trunk Verification

Objective: Confirm that trunk links are correctly established and carrying the expected VLANs.

```
show interfaces trunk
```

```
SW7#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Po2       on        802.1q         trunking    95
Po3       on        802.1q         trunking    95
Gig0/1    on        802.1q         trunking    95

Port      Vlans allowed on trunk
Po2       50,60,70,80,95
Po3       50,60,70,80,95
Gig0/1    50,60,70,80,95

Port      Vlans allowed and active in management domain
Po2       50,60,70,80,95
Po3       50,60,70,80,95
Gig0/1    50,60,70,80,95

Port      Vlans in spanning tree forwarding state and not pruned
Po2       50,60,70,80,95
Po3       50,60,70,80,95
Gig0/1    50,60,70,80,95
```

Figure 5 – HQ2 Trunk Configuration

12. Inter-VLAN Routing

The Multilayer Switch acts as the default gateway for every VLAN, enabling routing between VLANs at Layer 3.

VLAN 10 Network: 192.168.1.0/29

VLAN 10 Gateway: 192.168.1.1

VLAN 20 Network: 192.168.1.8/29

VLAN 20 Gateway: 192.168.1.9

VLAN 30 Network: 192.168.1.16/29

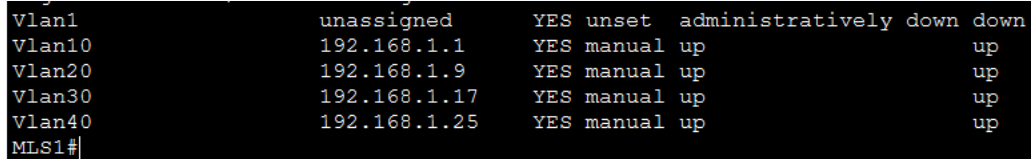
VLAN 30 Gateway: 192.168.1.17

VLAN 40 Network: 192.168.1.24/29

VLAN 40 Gateway: 192.168.1.25

Command:

```
show ip interface brief
```



Interface	IP Address	Status
Vlan1	unassigned	administratively down
Vlan10	192.168.1.1	up
Vlan20	192.168.1.9	up
Vlan30	192.168.1.17	up
Vlan40	192.168.1.25	up

Figure 6 – Multilayer Switch SVI Verification

13. Routing Design

Router	Routing
HQ1 Router	Default Static Route + PAT
HQ2 Router	OSPF + PAT
Core Router	OSPF
Server Router	OSPF + Static NAT
Bottom Router	OSPF + PAT

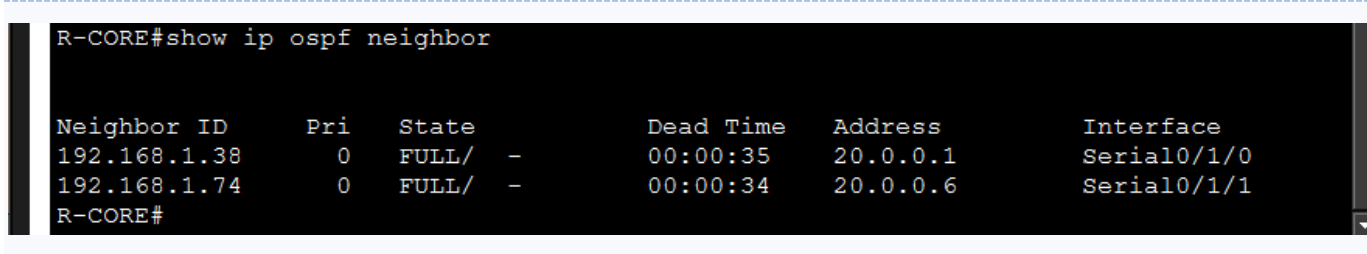
14. OSPF Configuration

OSPF is used to dynamically exchange routing information between the core router and the remote routers.

14.1 OSPF Neighbor Verification

Objective: Verify that OSPF neighbor relationships have been successfully established.

```
show ip ospf neighbor
```

A terminal window showing the output of the 'show ip ospf neighbor' command on a router named R-CORE. The output is a table with columns: Neighbor ID, Pri, State, Dead Time, Address, and Interface. Two neighbors are listed: 192.168.1.38 on Serial0/1/0 and 192.168.1.74 on Serial0/1/1, both in a FULL state.

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.38	0	FULL/ -	00:00:35	20.0.0.1	Serial0/1/0
192.168.1.74	0	FULL/ -	00:00:34	20.0.0.6	Serial0/1/1

Figure 7 – OSPF Neighbor Relationships

Result: The Core Router successfully established OSPF neighbor relationships with the required remote routers.

15. Routing Table Verification

Objective: Confirm that OSPF and static routes are correctly installed in the routing table.

```
show ip route
```

```
R-CORE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.1 to network 0.0.0.0

    20.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       20.0.0.0/30 is directly connected, Serial0/1/0
L       20.0.0.2/32 is directly connected, Serial0/1/0
C       20.0.0.4/30 is directly connected, Serial0/1/1
L       20.0.0.5/32 is directly connected, Serial0/1/1
C       20.0.0.8/30 is directly connected, Serial0/2/0
L       20.0.0.9/32 is directly connected, Serial0/2/0
C       20.0.0.12/30 is directly connected, Serial0/2/1
L       20.0.0.13/32 is directly connected, Serial0/2/1
    192.168.1.0/24 is variably subnetted, 11 subnets, 2 masks
O       192.168.1.0/29 [110/130] via 20.0.0.1, 00:24:09, Serial0/1/0
O       192.168.1.8/29 [110/130] via 20.0.0.1, 00:24:09, Serial0/1/0
O       192.168.1.16/29 [110/130] via 20.0.0.1, 00:24:09, Serial0/1/0
O       192.168.1.24/29 [110/130] via 20.0.0.1, 00:24:09, Serial0/1/0
O       192.168.1.32/30 [110/129] via 20.0.0.1, 00:24:09, Serial0/1/0
O       192.168.1.36/30 [110/128] via 20.0.0.1, 00:24:44, Serial0/1/0
O       192.168.1.40/29 [110/129] via 20.0.0.6, 00:24:44, Serial0/1/1
O       192.168.1.48/29 [110/129] via 20.0.0.6, 00:24:44, Serial0/1/1
O       192.168.1.56/29 [110/129] via 20.0.0.6, 00:24:44, Serial0/1/1
O       192.168.1.64/29 [110/129] via 20.0.0.6, 00:24:44, Serial0/1/1
O       192.168.1.72/30 [110/128] via 20.0.0.6, 00:24:44, Serial0/1/1
    198.168.1.0/32 is subnetted, 2 subnets
S       198.168.1.10/32 is directly connected, Serial0/2/0
S       198.168.1.11/32 is directly connected, Serial0/2/0
O*E2   0.0.0.0/0 [110/1] via 20.0.0.1, 00:24:44, Serial0/1/0
        [110/1] via 20.0.0.6, 00:24:44, Serial0/1/1
```

Figure 8 – Core Routing Table

The routing table on the Core Router is expected to show a mix of route types, indicated by the following codes:

Code	Meaning
O	OSPF intra-area route
O E2	OSPF external type-2 route

16. NAT & PAT Design

Site	NAT Type
HQ1	PAT
HQ2	PAT
Bottom	PAT
Server Site	Static NAT

17. Static NAT

Server	Private IP	Public IP
Server 1	192.168.3.5	198.168.1.10
Server 2	192.168.3.6	198.168.1.11

Static NAT maps each internal server to a dedicated public IP address.

```
show ip nat translations
```

```
192.168.3.5 <-> 198.168.1.10
```

```
192.168.3.6 <-> 198.168.1.11
```

```
R-UP#show ip nat translations
Pro  Inside global    Inside local      Outside local     Outside global
---  198.168.1.10      192.168.3.5      ---              ---
---  198.168.1.11      192.168.3.6      ---              ---
R-UP#
```

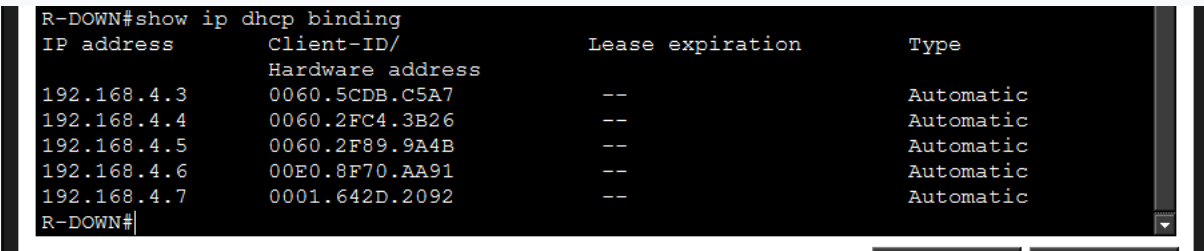
Figure 9 – Static NAT Mappings for Servers

18. DHCP Configuration

DHCP is configured on the Bottom Router to dynamically assign IP addresses to all client devices.

18.1 DHCP Bindings

```
show ip dhcp binding
```



IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.4.3	0060.5CDB.C5A7	--	Automatic
192.168.4.4	0060.2FC4.3B26	--	Automatic
192.168.4.5	0060.2F89.9A4B	--	Automatic
192.168.4.6	00E0.8F70.AA91	--	Automatic
192.168.4.7	0001.642D.2092	--	Automatic

Figure 10 – DHCP Address Bindings

18.2 DHCP Client Configuration

Path: PC → Desktop → IP Configuration → DHCP

IP Configuration



DHCP



Static

IPv4 Address

192.168.4.7

Subnet Mask

255.255.255.240

Default Gateway

192.168.4.1

DNS Server

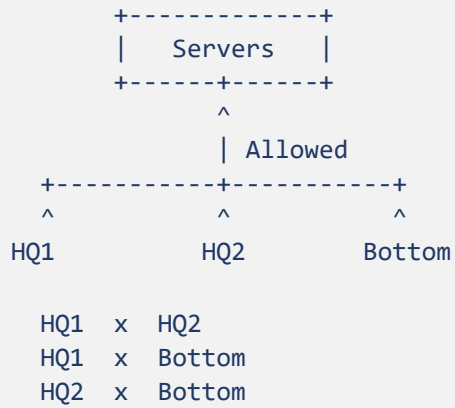
8.8.8.8

Figure 11 – DHCP Client Configuration

19. ACL & Network Security

The central security requirement of this project is:

Sites can access the servers but cannot communicate with each other.



20. Traffic Isolation

Allowed

- HQ1 → Server 1: Allowed
- HQ1 → Server 2: Allowed
- HQ2 → Server 1: Allowed
- HQ2 → Server 2: Allowed
- Bottom → Server 1: Allowed
- Bottom → Server 2: Allowed
- Same Site → Same Site: Allowed

Blocked

- HQ1 → HQ2: Blocked
- HQ1 → Bottom: Blocked
- HQ2 → HQ1: Blocked
- HQ2 → Bottom: Blocked
- Bottom → HQ1: Blocked
- Bottom → HQ2: Blocked

21. Connectivity Testing

Source	Destination	Expected	Result
HQ1 PC	Server 1	Success	Success
HQ1 PC	Server 2	Success	Success
HQ2 PC	Server 1	Success	Success
HQ2 PC	Server 2	Success	Success
Bottom PC	Server 1	Success	Success
Bottom PC	Server 2	Success	Success
HQ1 PC	HQ2	Blocked	Blocked
HQ1 PC	Bottom	Blocked	Blocked
HQ2 PC	Bottom	Blocked	Blocked

22. Ping Verification

22.1 HQ1 → Server 1

```
ping 198.168.1.10
```

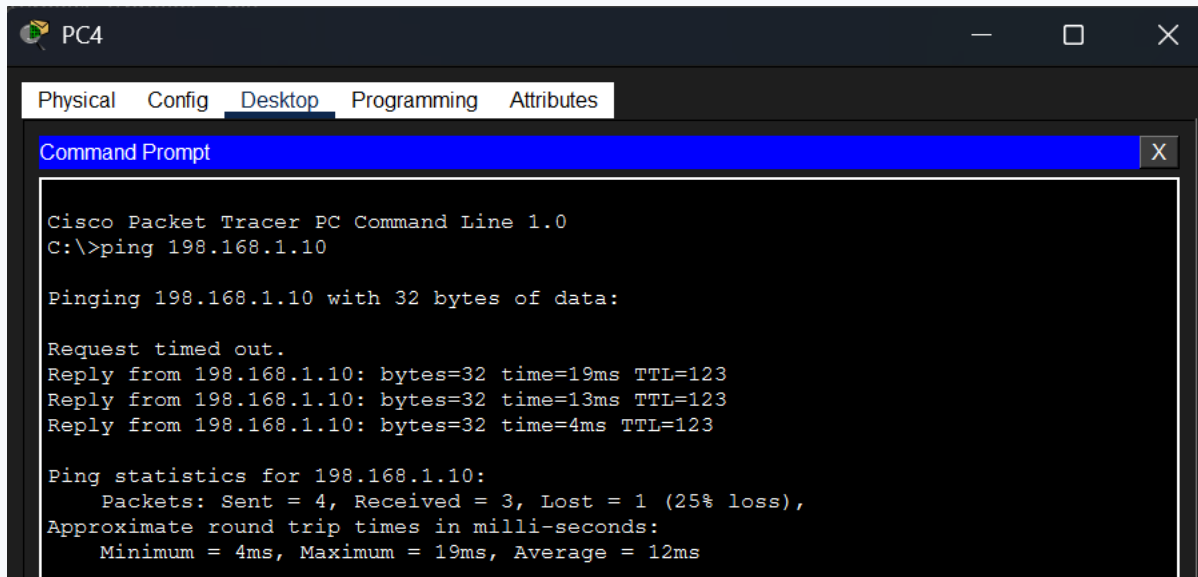


Figure 12 – HQ1 to Server 1 Ping

22.2 HQ1 → Server 2

```
ping 198.168.1.11
```

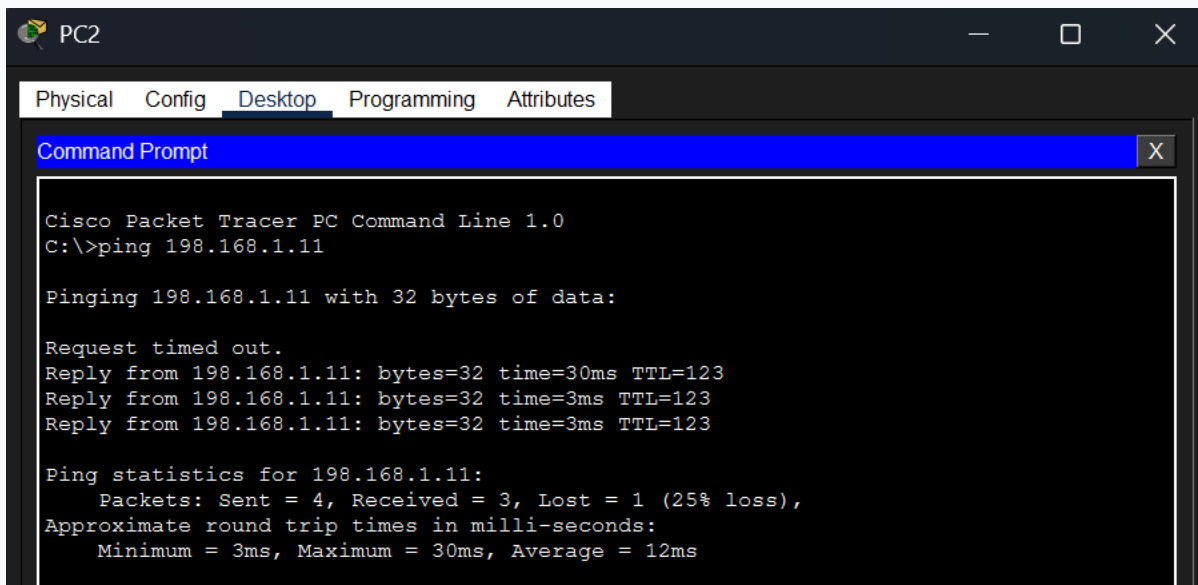
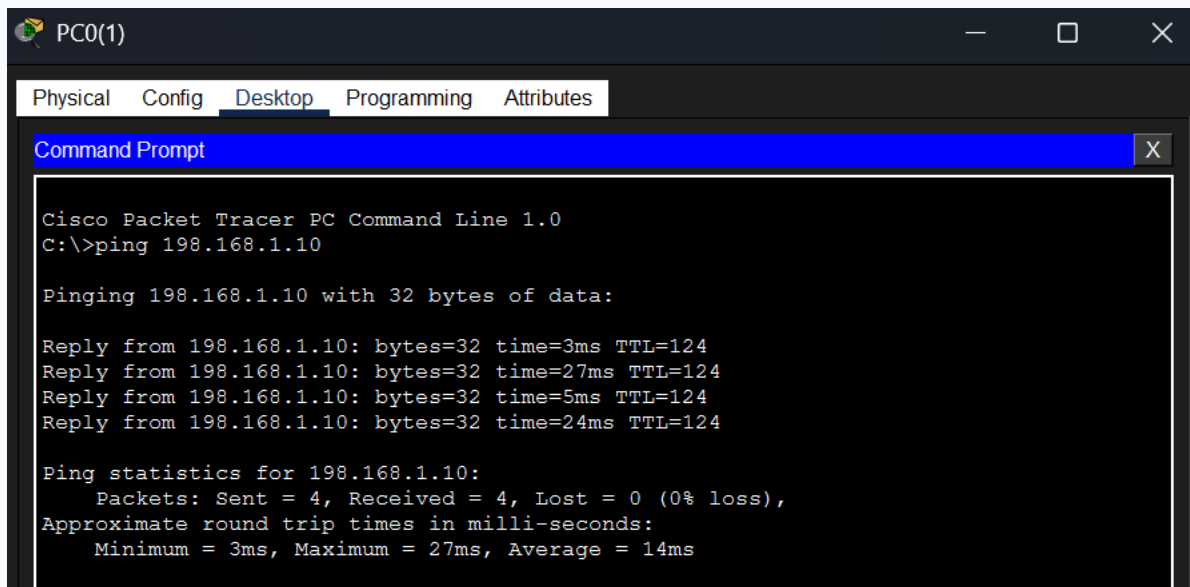


Figure 13 – HQ1 to Server 2 Ping

22.3 HQ2 → Server 1

```
ping 198.168.1.10
```



The screenshot shows a Cisco Packet Tracer PC window for PC0(1). The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of a ping command to 198.168.1.10. The output shows four successful replies with varying times (3ms, 27ms, 5ms, 24ms) and a TTL of 124. The ping statistics show 4 packets sent, 4 received, and 0% loss, with an average round trip time of 14ms.

```
PC0(1)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.168.1.10

Pinging 198.168.1.10 with 32 bytes of data:

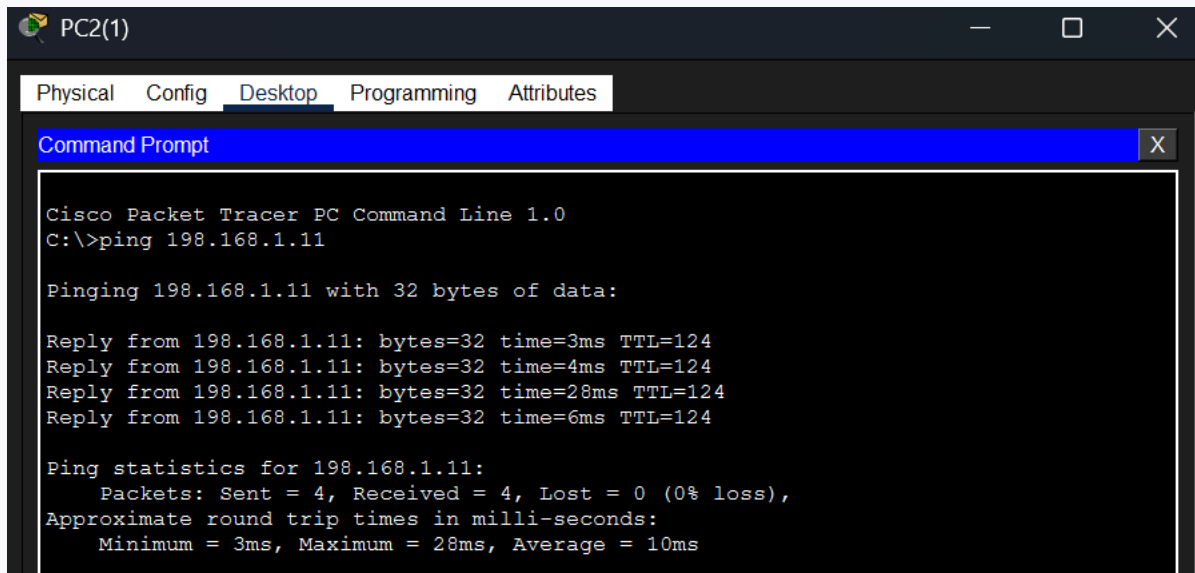
Reply from 198.168.1.10: bytes=32 time=3ms TTL=124
Reply from 198.168.1.10: bytes=32 time=27ms TTL=124
Reply from 198.168.1.10: bytes=32 time=5ms TTL=124
Reply from 198.168.1.10: bytes=32 time=24ms TTL=124

Ping statistics for 198.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 27ms, Average = 14ms
```

Figure 14 – HQ2 to Server 1 Ping

22.4 HQ2 → Server 2

```
ping 198.168.1.11
```



The screenshot shows a Cisco Packet Tracer PC window for PC2(1). The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the output of a ping command to 198.168.1.11. The output shows four successful replies with varying times (3ms, 4ms, 28ms, 6ms) and a TTL of 124. The ping statistics show 4 packets sent, 4 received, and 0% loss, with an average round trip time of 10ms.

```
PC2(1)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.168.1.11

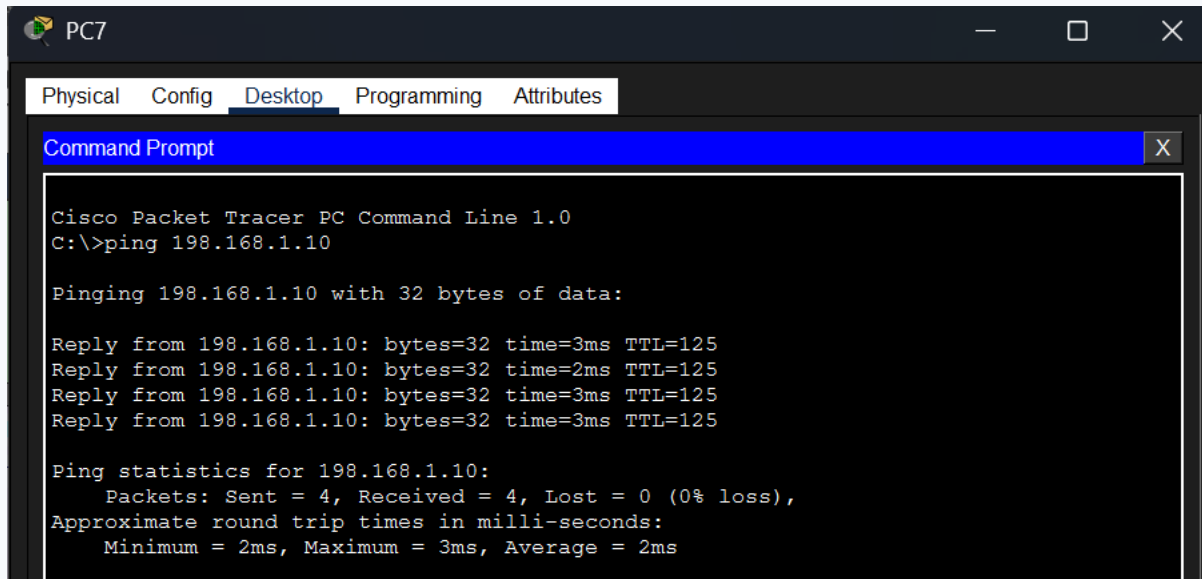
Pinging 198.168.1.11 with 32 bytes of data:

Reply from 198.168.1.11: bytes=32 time=3ms TTL=124
Reply from 198.168.1.11: bytes=32 time=4ms TTL=124
Reply from 198.168.1.11: bytes=32 time=28ms TTL=124
Reply from 198.168.1.11: bytes=32 time=6ms TTL=124

Ping statistics for 198.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 28ms, Average = 10ms
```

Figure 15 – HQ2 to Server 2 Ping

22.5 Bottom → Server 1



The screenshot shows a window titled 'PC7' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the output of a ping command to 198.168.1.10. The output indicates successful connectivity with 0% packet loss and an average round-trip time of 2ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.168.1.10

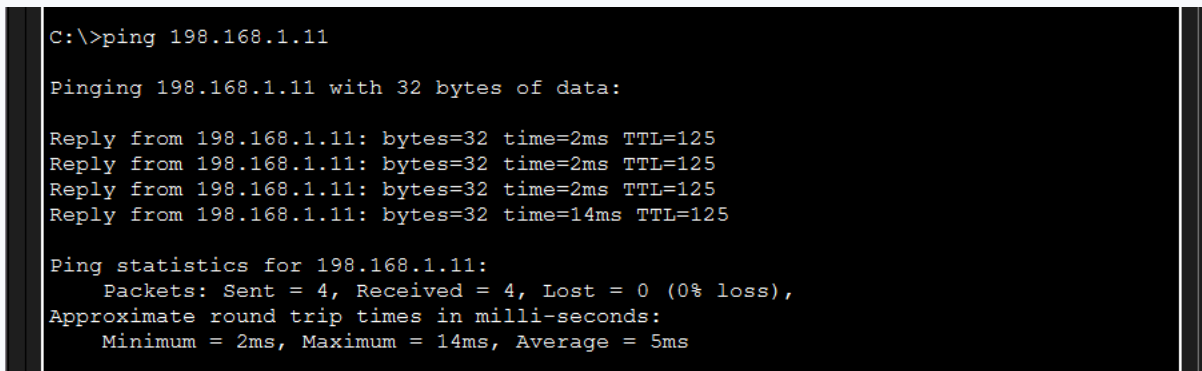
Pinging 198.168.1.10 with 32 bytes of data:

Reply from 198.168.1.10: bytes=32 time=3ms TTL=125
Reply from 198.168.1.10: bytes=32 time=2ms TTL=125
Reply from 198.168.1.10: bytes=32 time=3ms TTL=125
Reply from 198.168.1.10: bytes=32 time=3ms TTL=125

Ping statistics for 198.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

Figure 16 – Bottom to Server 1 Ping

22.6 Bottom → Server 2



The screenshot shows a Command Prompt window with the output of a ping command to 198.168.1.11. The output indicates successful connectivity with 0% packet loss and an average round-trip time of 5ms.

```
C:\>ping 198.168.1.11

Pinging 198.168.1.11 with 32 bytes of data:

Reply from 198.168.1.11: bytes=32 time=2ms TTL=125
Reply from 198.168.1.11: bytes=32 time=2ms TTL=125
Reply from 198.168.1.11: bytes=32 time=2ms TTL=125
Reply from 198.168.1.11: bytes=32 time=14ms TTL=125

Ping statistics for 198.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 14ms, Average = 5ms
```

Figure 17– Bottom to Server 2 Ping

23. Security Testing

The following tests confirm that the isolation policy is correctly enforced — each of these pings is expected to fail (Request Timeout).

```
HQ1 PC
  |
  v
HQ2 Network
  |
  v
Request Timeout
```

23.1 HQ1 → HQ2

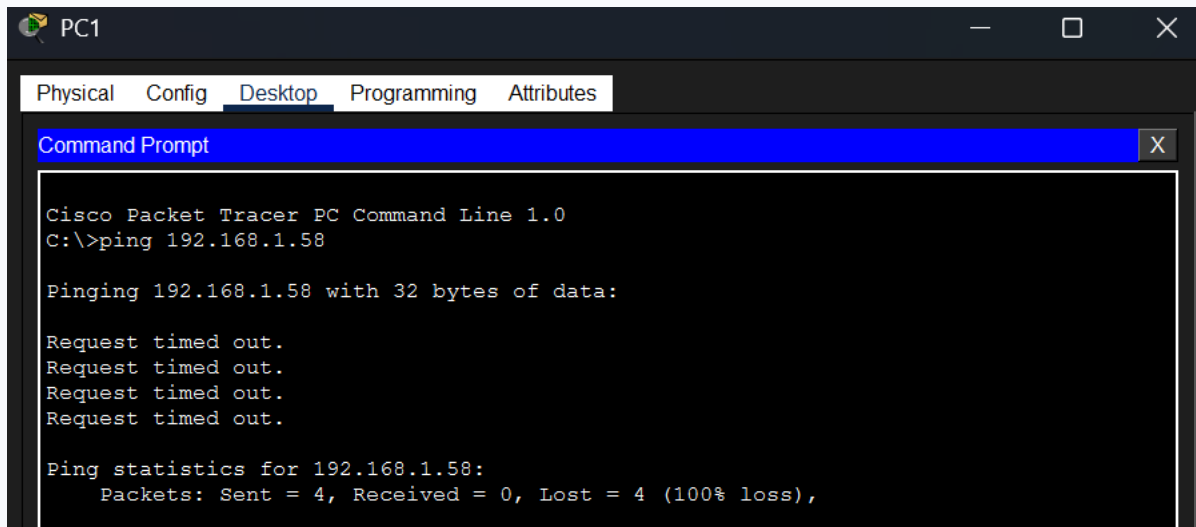


Figure 18 – HQ1 to HQ2 Isolation Test

23.2 HQ1 → Bottom

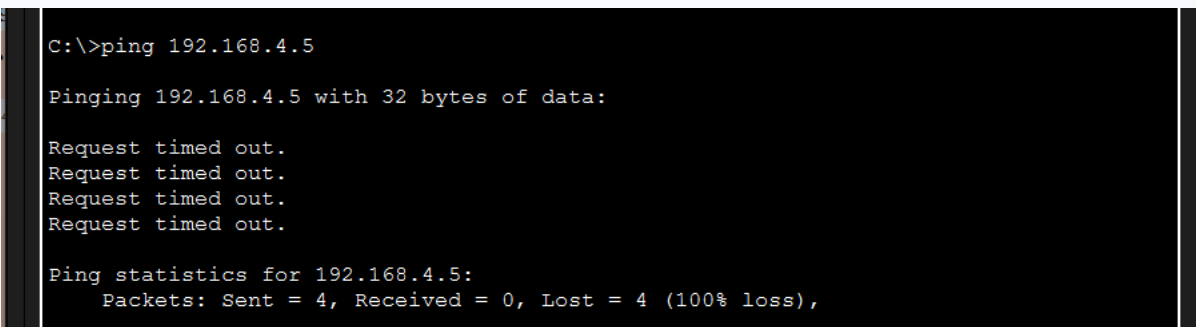


Figure 19 – HQ1 to Bottom Isolation Test

23.3 HQ2 → Bottom

```

C:\>ping 192.168.4.7

Pinging 192.168.4.7 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.4.7:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure 20 – HQ2 to Bottom Isolation Test

These results confirm that the security requirement for inter-site isolation has been fully met.

24. Console & Enable Password Security

On every device local Console access and the Privileged EXEC (Enable) password were secured, with password encryption enabled for the stored configuration.

```

User Access Verification

Password:

R-CORE>en
Password:
R-CORE#

```

25.Remote Access — Telnet & SSH

Telnet (Basic remote access)

```

Device(config)# line vty 0 15
Device(config-line)# password *****
Device(config-line)# login local
Device(config-line)# transport input all
Device(config-line)# exit

```

Note: SSH was enabled on all managed routers and switches, and Telnet was disabled on sensitive ports in favor of transport input ssh to ensure secure remote access. Access via PuTTY / Terminal from a PC using the management IP was tested successfully.

26. Final Network Validation

Infrastructure

- VLANs configured
- Trunks configured
- Inter-VLAN routing working
- IP addressing completed
- WAN links working

Routing

- OSPF neighbors established
- Static default route configured
- Server routes reachable

NAT

- HQ1 PAT working
- HQ2 PAT working
- Bottom PAT working
- Server Static NAT working

Services

- DHCP working
- Servers reachable

Security

- Same-site communication working
- Server access allowed
- Site-to-site communication blocked
- SSH-TELNET
- Enable&Login Passwords

27. Conclusion

The project successfully implements a multi-site enterprise network using Cisco Packet Tracer. VLAN segmentation and subnetting were used to organize the internal networks, while OSPF and static routing were implemented to provide controlled connectivity across the WAN.

NAT and PAT were configured to provide public access to the required resources while maintaining private addressing internally. DHCP was implemented for automatic IP address assignment at the Bottom site.

Finally, traffic isolation was implemented to ensure that each site can communicate with the centralized servers and with its own internal devices, while direct communication between different sites is restricted.

The final network was verified using routing, NAT, DHCP, OSPF, and end-to-end connectivity tests.